



University of
Nottingham
UK | CHINA | MALAYSIA

COMP3055

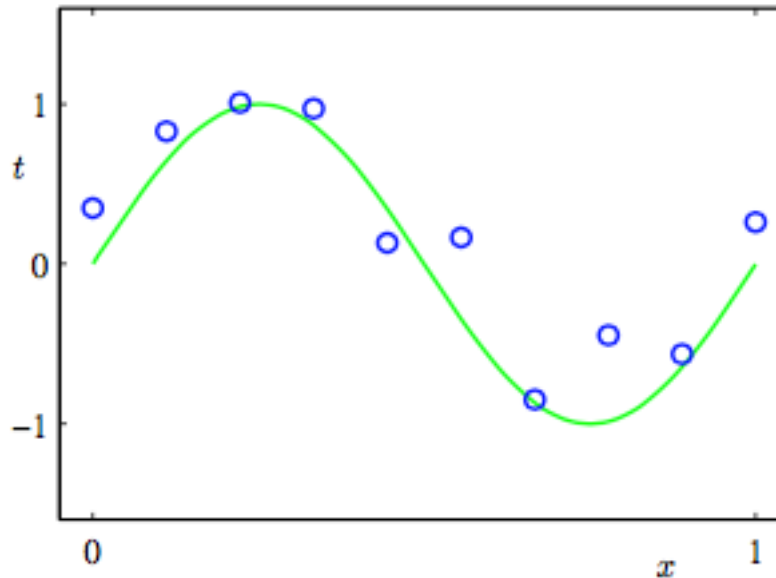
Machine Learning

Topic 4 – Machine Learning Theory and Practice

Prof. Ying Weng
2025 Autumn

Motivating Example

Polynomial Curve Fitting 多项式曲线拟合



Plot of a training data set of $N = 10$ points, each comprising an observation of the input variable x along with the corresponding target variable t . The green curve shows the function $\sin(2\pi x)$ used to generate the data. Our goal is to predict the value of t for some new value of x , **without** knowledge of the green curve.

Classification VS Regression

Classification 分类预测离散
predict a label (**discrete value**)

Regression 回归预测连续
predict a response (**continuous value**)

Regression Problem

- $X \equiv (x_1, \dots, x_N)^T$, $T \equiv (t_1, \dots, t_N)^T$
- **Training set** is generated with $t_n = \sin(2\pi x_n) + b_n$, $n = 1, 2, \dots, N$, where b_n is **random noise** having a Gaussian Distribution.
- **Goal**: predict the target value of t for some new input value $x \rightarrow$ implicitly trying to discover the underlying function $\sin(2\pi x_n)$.
- Generalize from a finite data set ($N=10$)
- **Uncertainty**: the observed data are corrupted with noise.

Polynomial Function

- We fit the training data using a **polynomial function** (linear models) of the form:

$$y(x, \mathbf{w}) = w_0 + w_1 x^1 + \dots + w_M x^M = \sum_{j=0}^M w_j x^j$$

- Where M is the 阶数=模型自由度 **order of the polynomial** (degree of freedom), and the **polynomial coefficients** $w_0, w_1, \dots, w_M$ are collectively denoted by the vector $\mathbf{w}$ w =多项式的系数

Solve a Polynomial Function

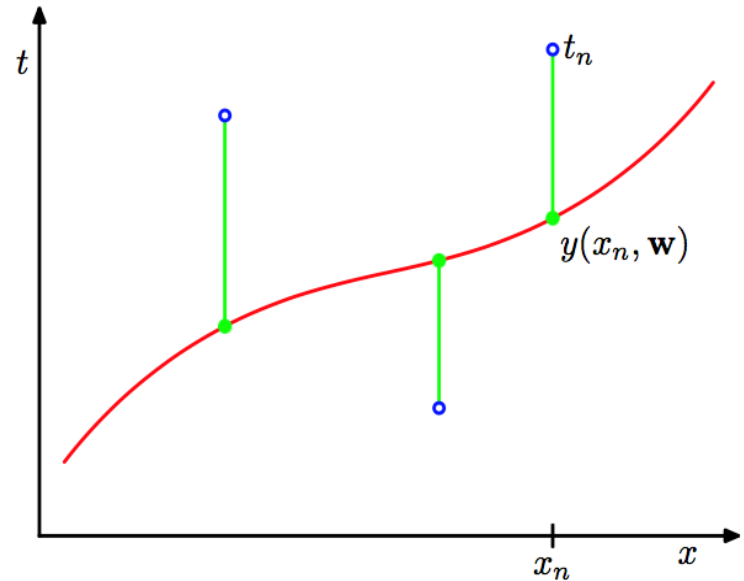
Target: find the values of *polynomial coefficients*

- Step 1: Fit the polynomial to the training data
- Step 2: minimize the *error function*

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

Where $E(\mathbf{w})$ measures the misfit between the function $y(x, \mathbf{w})$ and the training set data points

- 蓝点: 训练数据点 (x_n, t_n)
- 红线: 模型预测函数 $y(x, \mathbf{w})$
- 绿色线段: 误差 (残差 residual) → 模型预测与真实值的距离

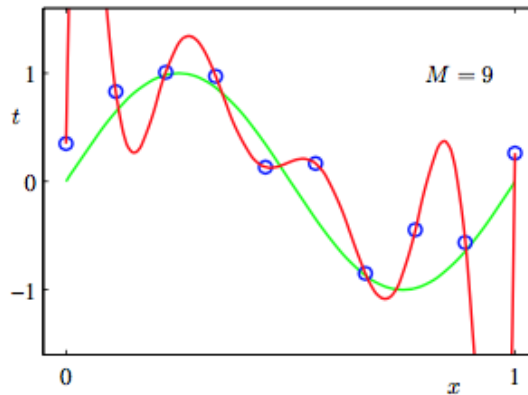
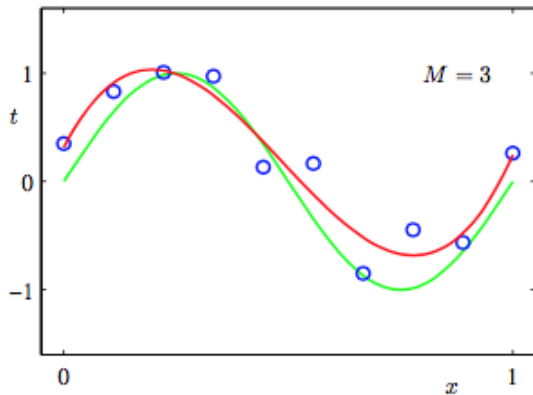
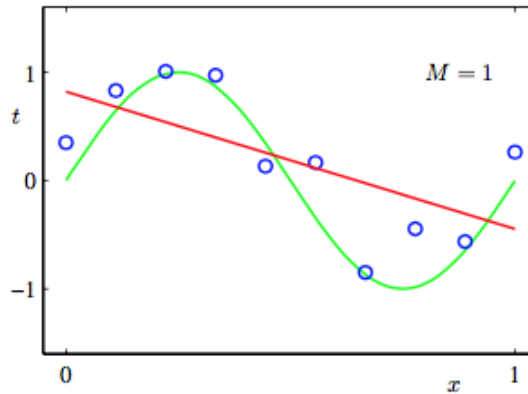
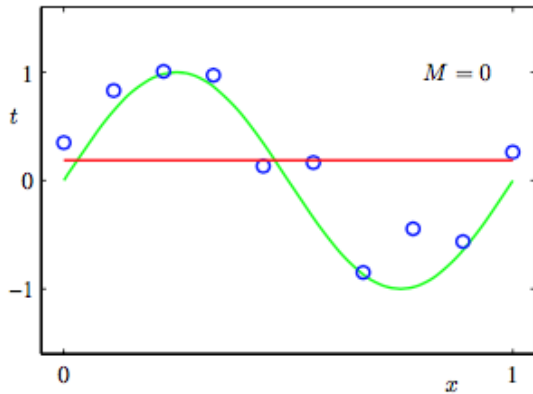


- 衡量模型预测 $y(x_n, \mathbf{w})$ 与真实值 t_n 之间的偏差。
- $\frac{1}{2}$ 是数学上方便推导 (求导时系数消失)。
- 若 $E(\mathbf{w})=0$, 说明模型完全穿过所有样本点 (通常过拟合)。
- 因为 $E(\mathbf{w})$ 是关于 $\mathbf{w}$ 的二次函数, 它只有一个唯一最优解。

$\frac{1}{2}$ is included for later convenience. $E(\mathbf{w})=0$ if and only if the function $y(x, \mathbf{w})$ were to pass exactly through each training data point. The error function is a quadratic function of $\mathbf{w}$, there is only one solution.

Model Selection

Choosing order M of the polynomial



阶数 M	拟合情况	特征
$M=0$	常数函数 (红线水平)	欠拟合 (underfitting), 过于简单, 不能捕捉趋势
$M=1$	直线拟合	仍然欠拟合, 趋势过于粗糙
$M=3$	三次多项式	逼近良好, 平滑且泛化性较好
$M=9$	九次多项式	严重过拟合, 穿过所有点但波动剧烈

模型复杂度 M 的选择是关键：太小 欠拟合，太大 过拟合。

Model Selection

- **Goal:** achieve good *generalization* by making accurate predictions for new data.

使用 均方根误差 (Root Mean Square Error, RMS) 衡量拟合效果：

- We use Root-mean-square (RMS) error on data

$$E_{RMS} = \sqrt{2E(w^*)/N}$$

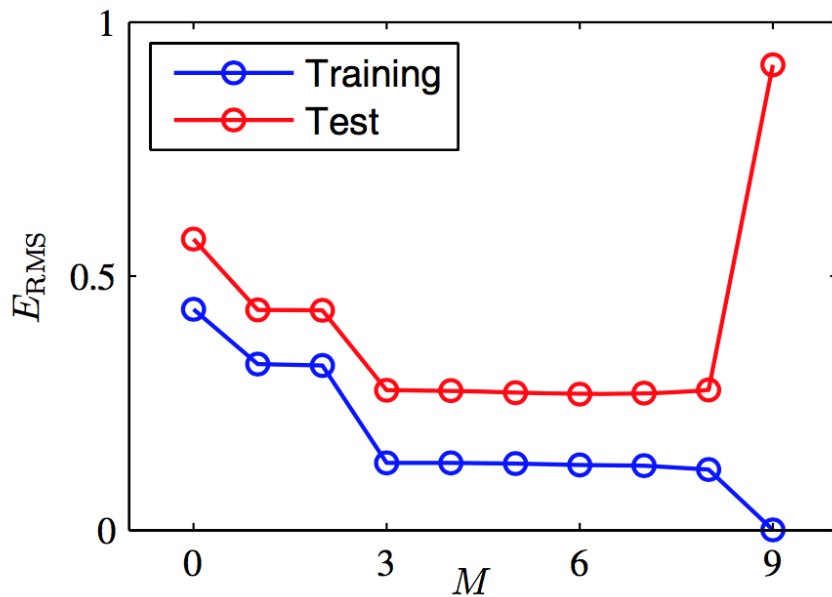
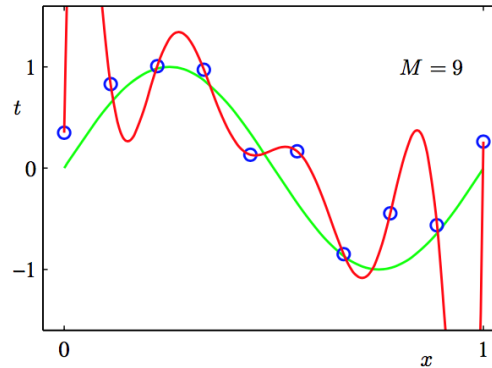
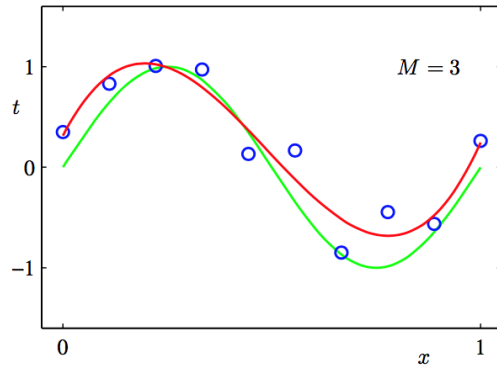
RMS 误差衡量模型预测值与真实值的平均偏差。

可以跨不同数据规模比较（因为除以 N 进行了归一化）。

RMS 越小 模型性能越好。

- Where N allows us to compare different sizes of data set, and w^* is the *solution* of minimizing $E(w)$ (*hypothesis*).
- It measures how well the model w^* doing in predicting the values of t for new data observations of x .

Overfitting



	$M = 0$	$M = 1$	$M = 6$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

Overfitting

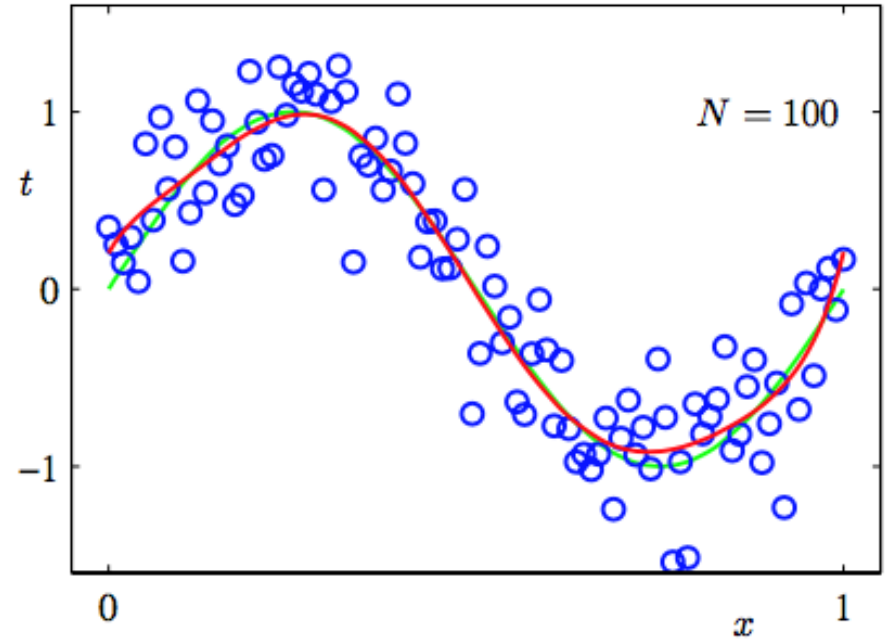
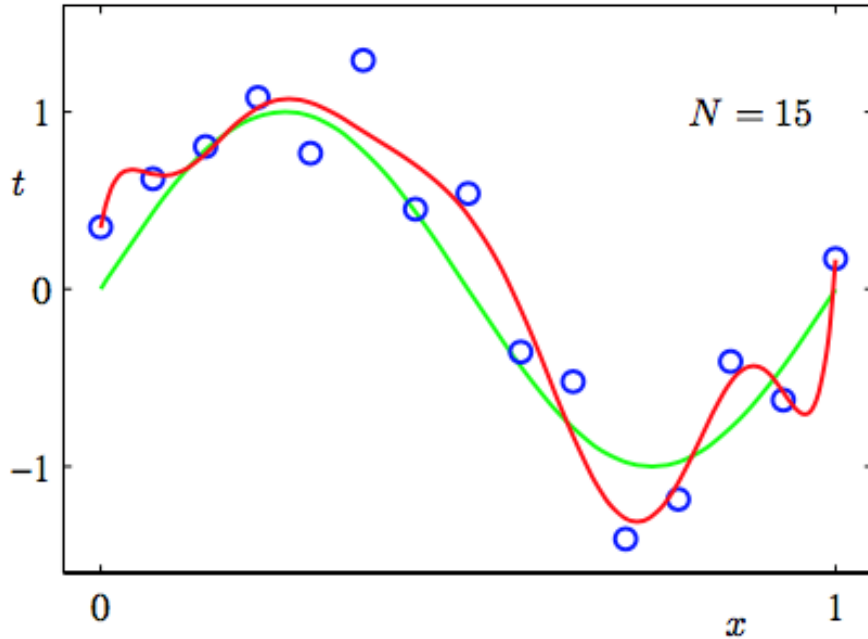
Overfitting can occur when:

学习时间太长

- Learning is performed for **too long** (e.g. in Neural Networks). 训练数据代表性不足 (Not representative) 训练集未能覆盖实际数据分布的全部情况。
- The examples in the training set are **not representative** of all possible situations (is usually the case!).
- Model parameters are adjusted to **uninformative features** in the training set that have no causal relation to the true underlying target function!

模型学习了无关特征 (Uninformative features) 模型把噪声或随机模式当作有用信息去拟合

Overfitting



Increasing the size of the data set reduces the overfitting problem.

一个简单但有效的解决办法：增加训练数据量。

Regularization

正则化通过在误差函数中加入“惩罚项 (penalty term)”，限制参数 w 的过大波动，从而防止过拟合。

- *Regularization* can control the overfitting phenomenon, by adding *penalty term* to the error function to discourage the coefficients from reaching large values.

$$E(w) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, w) - t_n\}^2 + \frac{\lambda}{2} \|w\|^2$$

- Where $\|w\|^2 \equiv w^T w = w_0^2 + w_1^2 + \dots + w_M^2$, and the coefficient λ governs the relative importance of the regularization term compared with the sum-of-squares error term.

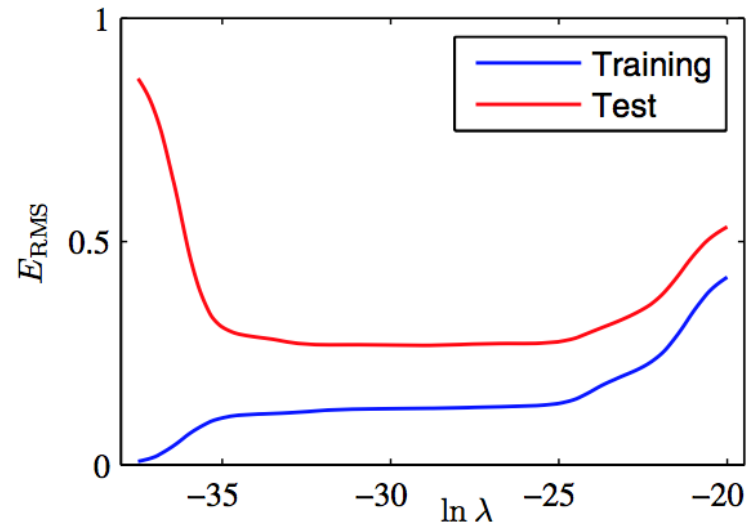
- 当 $\lambda = 0 \rightarrow$ 无正则化，模型可能会过拟合；
- 当 λ 太大 $\rightarrow$ 模型被过度约束，容易欠拟合；
- 适中的 $\lambda \rightarrow$ 在“拟合度”与“泛化能力”之间取得平衡。

■ 正则化的核心目标：

通过限制系数 w_j 的大小，减少模型对噪声的敏感性，提升泛化能力。

Regularization

	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
w_0^*	0.35	0.35	0.13
w_1^*	232.37	4.74	-0.05
w_2^*	-5321.83	-0.77	-0.06
w_3^*	48568.31	-31.97	-0.05
w_4^*	-231639.30	-3.89	-0.03
w_5^*	640042.26	55.28	-0.02
w_6^*	-1061800.52	41.32	-0.01
w_7^*	1042400.18	-45.95	-0.00
w_8^*	-557682.99	-91.53	0.00
w_9^*	125201.43	72.68	0.01

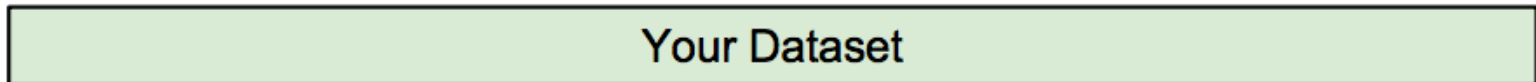


- Table of the coefficients w^* for $M = 9$ polynomials with various values for the regularization parameter λ .
- λ controls the *effective complexity* of the model and hence determines the *degree of overfitting*

Cross Validation

超参数

- **Idea #1:** Choose **hyperparameters** that work best on the data



- **Idea #2:** Split data into **train** and **test**, choose hyperparameters that work best on test data



- **Idea #3:** Split data into **train**, **val**, and **test**; choose hyperparameters on **val** and evaluate on **test**



三分法 (train/val/test) 是调参与评估模型的正确方式，防止模型对测试集产生“信息泄露”。

Cross Validation

- **Idea #4: Cross-Validation**: Split data into **folds**, try each fold as validation and average the results

fold 1	fold 2	fold 3	fold 4	fold 5	test
fold 1	fold 2	fold 3	fold 4	fold 5	test
fold 1	fold 2	fold 3	fold 4	fold 5	test
fold 1	fold 2	fold 3	fold 4	fold 5	test
fold 1	fold 2	fold 3	fold 4	fold 5	test

1. 将数据划分为 K 个折叠 (folds)

例如: $K=5 \rightarrow \text{fold}_1, \text{fold}_2, \text{fold}_3, \text{fold}_4, \text{fold}_5$

2. 每次选择一个 fold 作为验证集, 其余 $K-1$ 个作为训练集

→ 进行 K 次训练与验证。

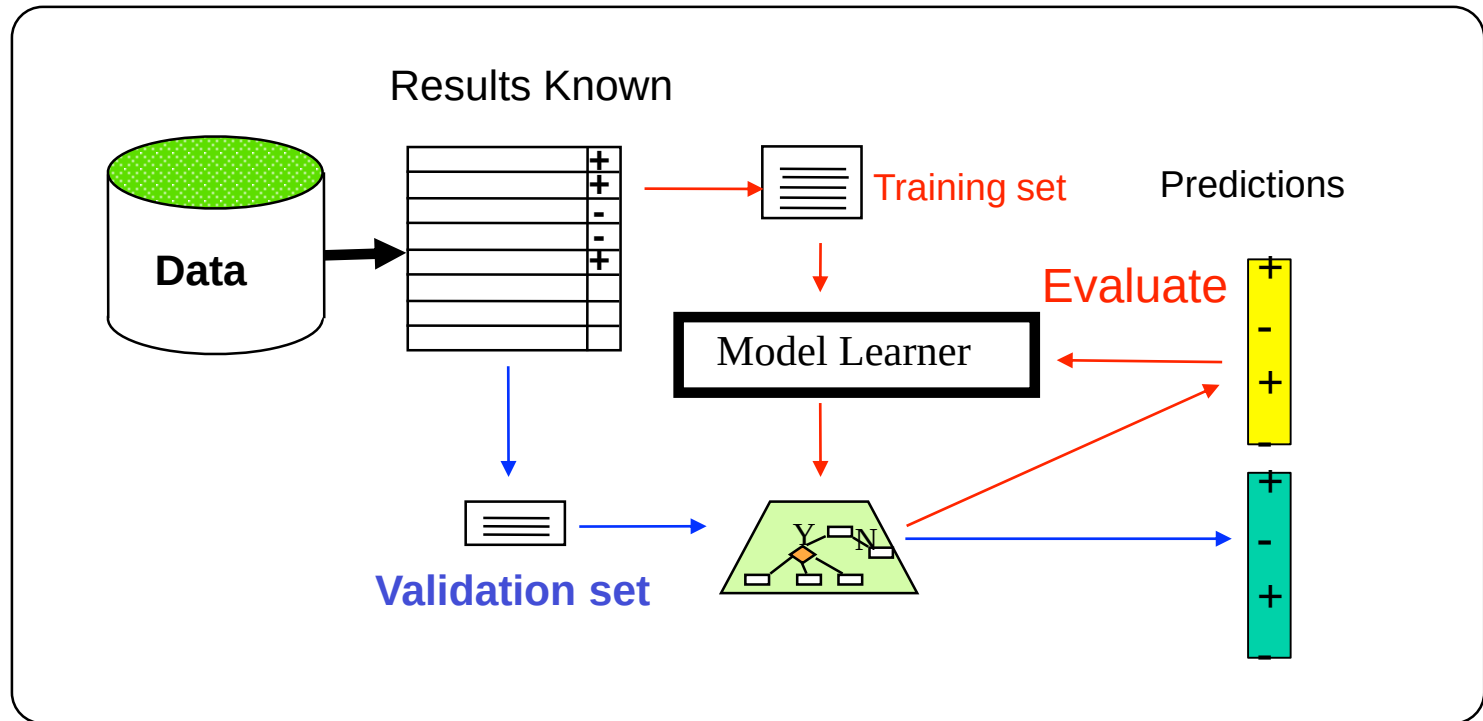
3. 记录每次的验证误差并取平均值

→ 得到更稳定、客观的性能评估。

Cross Validation

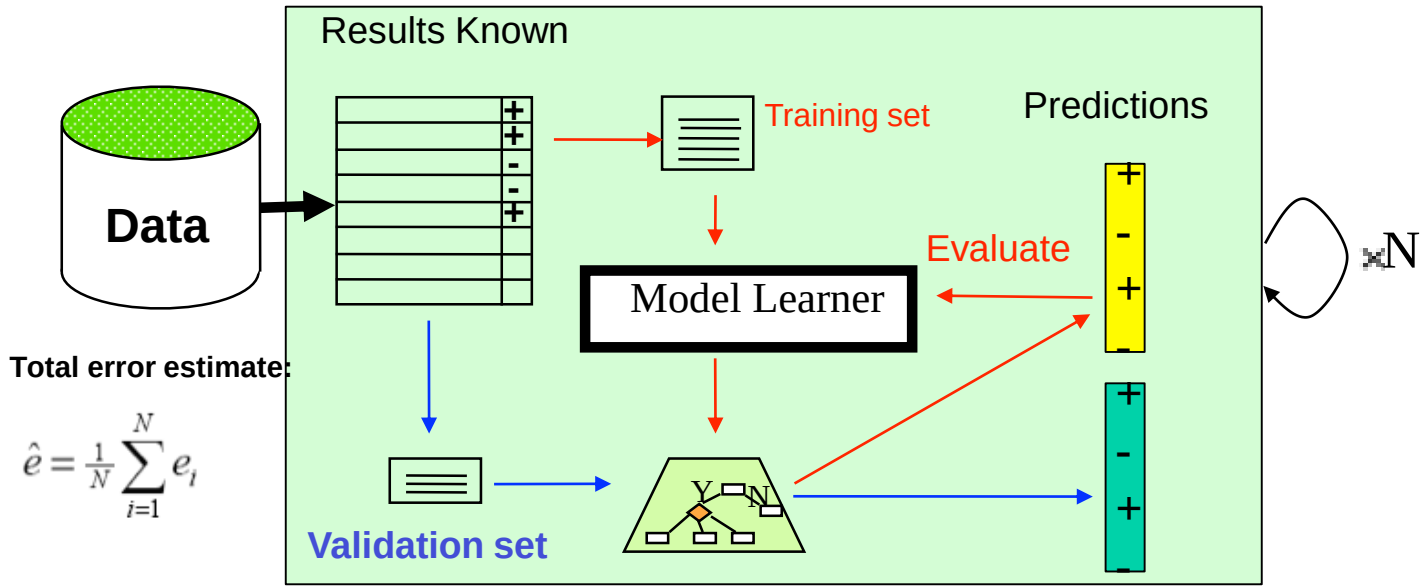
- **Cross Validation** is often used to counter overfitting.
- Partition the dataset into S groups, with $(S-1)$ training sets, a validation set and a testing set.
 - The training set is used to determine the coefficients w
 - The validation set is used to optimize the model complexity (hyperparameters, either M or λ in the previous example)
 - The testing set is used to evaluate the final selected model
- The procedure is then repeated for all S possible choices, the performance scores from the S runs are then averaged.

Evaluation procedure



- For large datasets, a single split is usually sufficient.
- For smaller datasets, rely on cross validation

Cross validation procedure



- Split the data into training, validation and test sets in a repeated fashion.
- Estimate the total error as the average of each fold error.

Classification Measures

• Error Rate (Accuracy)	Error Rate / Accuracy	错误率与准确率，用于总体判断模型预测正确的比例
• Precision/Recall	Precision / Recall	精确率与召回率，适用于类别不平衡的数据
• F-measure	F-measure (F1-score)	同时考虑 Precision 和 Recall 的综合指标
• ROC curve	ROC curve (Receiver Operating Characteristic)	衡量不同阈值下模型性能的曲线图
• Confusion matrix	Confusion matrix	直观显示预测类别与真实类别的对应关系
• ...	...	其他如 AUC、Specificity 等

Error Rate

- Common performance measure for classification problems
 - **Success**: instance's class is predicted correctly (True Positives (**TP**) / Negatives (**TN**))
 - Error: instance's class is predicted incorrectly (False Positives (**FP**) / Negatives (**FN**))
 - **False positives** - **Type I error**. False Negative - **Type II error**
- Classification **error rate**: proportion of instances misclassified over the whole set of instances
$$\text{Error Rate} = (\text{FP} + \text{FN}) / (\text{P} + \text{N})$$
a.k.a, **accuracy**
$$\text{Accuracy} = 1 - \text{Error Rate}$$
- Classification Error Rate on the Training Set can be too optimistic!

Unbalanced data

- Balanced set: (roughly) equal number of positive / negative examples:

Classifier	TP	TN	FP	FN	Recall Rate
A	25	25	25	25	50%
B	37	37	13	13	74%

Recall = $TP / (TP + FN)$ 召回率更高---> 能识别更多正样本

Unbalanced data

- Unbalanced set: unequal number of positive / negative examples

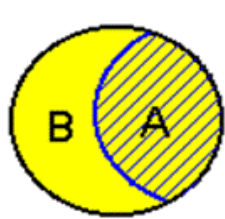
Classifier	TP	TN	FP	FN	Recall Rate
A	25	75	75	25	50%
B	0	150	0	50	0%

无法识别任何正样本---> Recall = 0 ---> “无效分类器”。

Classifier B cannot predict any positive examples!

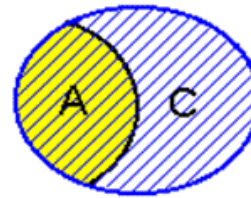
$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Recall/Precision



B: Number of positive examples not retrieved
A: Number of positive examples retrieved

$$\text{RECALL: } \frac{A}{A+B} \times 100\%$$



C: Number of negative examples retrieved
A: Number of positive examples retrieved

$$\text{PRECISION: } \frac{A}{A+C} \times 100\%$$

Recall (召回率)

能找出的正样本比例

$$\frac{TP}{TP+FN}$$

衡量“找全”能力

Precision (精确率)

预测为正样本中真正正确的比例

$$\frac{TP}{TP+FP}$$

衡量“找准”能力

More insight over a classifier's behavior

For the positive class:

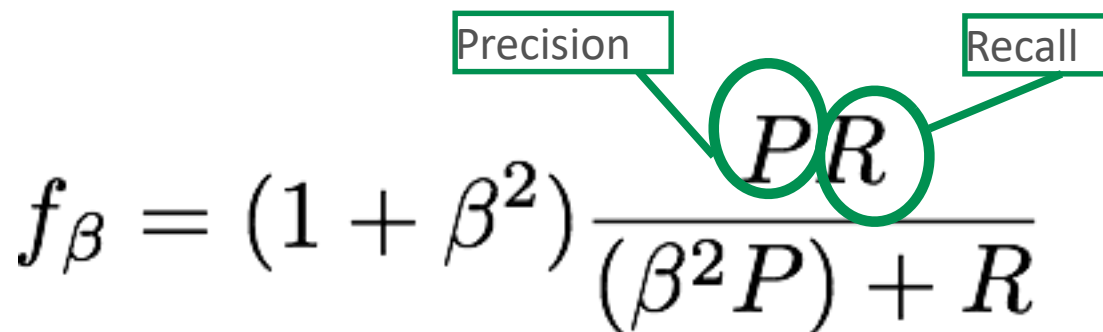
Classifier A: Recall = 50%, Precision = 25%

Classifier B: Recall = 0%, Precision = 0%

Classifier B is useless!

F-Measure (also called as F-Score)

- Comparing different approaches is difficult when using multiple evaluation measures (e.g. Recall and Precision)
- F-Measure (also called as F-Score) combines recall and precision into a single measure:



The diagram shows the formula for the F-beta score: $f_{\beta} = (1 + \beta^2) \frac{PR}{(\beta^2 P) + R}$. The letters 'P' and 'R' in the numerator are circled in green. A green box labeled 'Precision' has a line pointing to the 'P', and another green box labeled 'Recall' has a line pointing to the 'R'.

$$f_{\beta} = (1 + \beta^2) \frac{PR}{(\beta^2 P) + R}$$

β is a non-negative real values

F1 越高，说明模型在 Precision 与 Recall 上的平衡越好。
当数据不平衡时，比 Accuracy 更可靠。

- We often use F1-Measure (also called as F1-Score).

F-Measure (also called as F-Score)


$$f_{\beta} = (1 + \beta^2) \frac{PR}{(\beta^2 P) + R}$$

if $\beta=0.5$, then $f_{\beta} = ?$ For P and R , which is more important?

if $\beta=1$, then $f_{\beta} = ?$ For P and R , which is more important?

if $\beta=2$, then $f_{\beta} = ?$ For P and R , which is more important?

F-Measure (also called as F-Score)


$$f_{\beta} = (1 + \beta^2) \frac{PR}{(\beta^2 P) + R}$$

if $\beta=0.5$, then $f_{\beta} = ?$ For P and R , which is more important?

if $\beta=0.5$, then $f_{0.5} = 1.25PR / (0.25P+R)$

For P and R , P is more important than R

if $\beta=1$, then $f_{\beta} = ?$ For P and R , which is more important?

if $\beta=1$, then $f_1 = 2PR / (P+R)$

For P and R , P is as important as R

if $\beta=2$, then $f_{\beta} = ?$ For P and R , which is more important?

if $\beta=2$, then $f_2 = 5PR / (4P+R)$

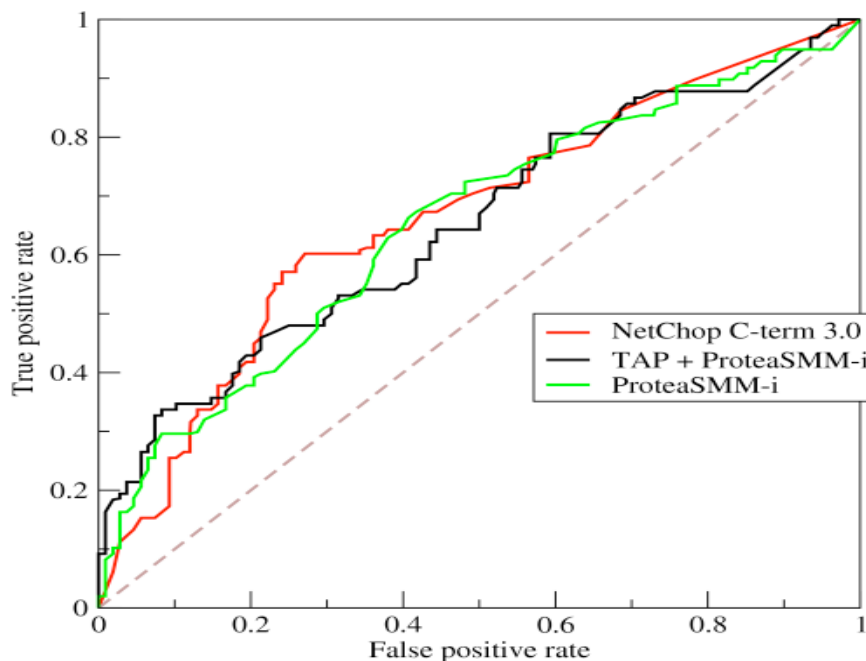
For P and R , R is more important than P

不同阈值下的 真正例率 (TPR) 与 假正例率 (FPR) 之间的关系。

ROC curves

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

- Receiver Operator Characteristic (ROC) curves plot TP vs FP rates



曲线特点:

- 横轴: FPR (False Positive Rate)
- 纵轴: TPR (True Positive Rate)
- 曲线越靠近左上角 → 模型性能越好。

AUC (Area Under Curve)

- 曲线下面积 (AUC) 越大 → 模型越优秀。
- 随着决策阈值变化, 可观察模型的稳定性。

- Can be achieved by e.g. varying decision threshold of a classifier
- Area under the curve is often used as measure of goodness

Confusion matrix

混淆矩阵：一种用于展示模型分类结果的可视化表格。

- A visualization tool used to present the results attained by a learner.
- Easy to see if the system is commonly mislabeling one class as another.

Prediction \ True	A	B	C
A	5	3	0
B	2	3	1
C	0	2	11

当特征维度 D 增大时：
模型参数数目呈指数增长；
需要的数据量也急剧上升；
学习变得困难。

Curse of Dimensionality

- As dimension D increases, the number of **independent coefficients** grows proportionally

$$y(x, \mathbf{w}) = w_0 + w_1 x^1 + \cdots + w_M x^M = \sum_{j=0}^M w_j x^j$$


$$y(x, \mathbf{w}) = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D \sum_{j=1}^D w_{ij} x_i x_j + \sum_{i=1}^D \sum_{j=1}^D \sum_{k=1}^D w_{ijk} x_i x_j x_k \dots$$

- The **amount of data** needed to support the result often **grows exponentially** with the dimensionality.

Curse of Dimensionality

- Consider a sphere of radius $r = 1$ in a space of D dimensions, what is the fraction of the volume of the sphere that lies between radius $r = 1 - \epsilon$ and $r = 1$?
- The volume of a sphere of radius r in D dimensions must scale as r^D , we have

• 半径为 1 的 D 维球体:

$$V_D(r) = K_D r^D$$

$$V_3(r) = \frac{4}{3} \pi r^3$$

• 当维度 D 增大, 体积集中在外层, 数据越来越稀疏。

$$V_D(r) = K_D r^D$$

 直觉:

在高维空间里, 样本之间“几乎互不重叠”, 导致模型泛化性显著下降。

“维度灾难”是指随着特征数增加, 模型所需数据量与计算复杂度呈指数增长, 使学习与泛化变得困难。

Curse of Dimensionality

- Where the **constant** K_D depends only on D , thus the required volume fraction is given by

$$\frac{V_D(1) - V_D(1 - \epsilon)}{V_D(1)} = 1 - (1 - \epsilon)^D$$

- For large D , the fraction **tend to 1** even for small values of ϵ .

考虑单位球 $r = 1$, 我们想计算球体表面“外层薄壳”所占体积比例 (即半径在 $1 - \epsilon$ 与 1 之间的部分):

$$\text{Fraction} = \frac{V_D(1) - V_D(1 - \epsilon)}{V_D(1)}$$

代入 $V_D(r) = K_D r^D$ 得:

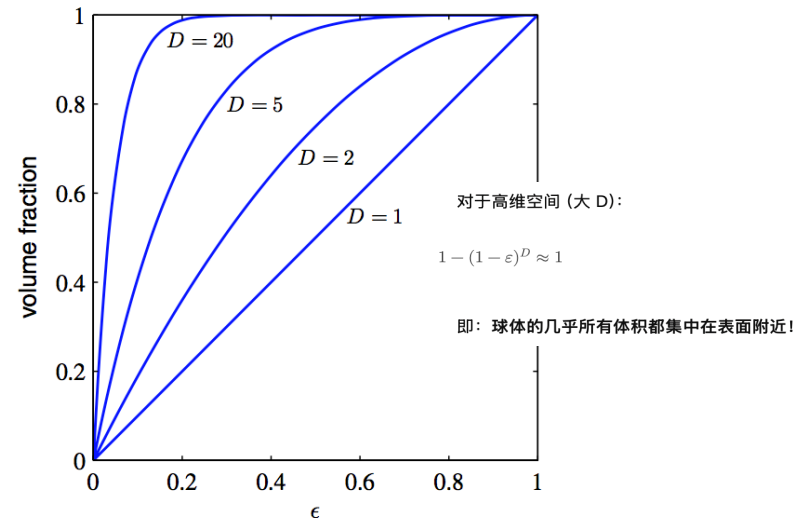
$$\text{Fraction} = 1 - (1 - \epsilon)^D$$

图中显示了当维度 $D = 1, 2, 5, 20$ 时的体积分布。

- 横轴: ϵ (靠近外层的厚度)
- 纵轴: 外层体积占比 (Volume Fraction)

从图可见:

- 当 $D = 1$ 时 (线性空间), 体积均匀分布;
- 随着 D 增大, 即使 ϵ 很小, 外层体积占比也迅速逼近 1。



Curse of Dimensionality

- For large D , the fraction *tend to 1* even for small values of ϵ .
- In space of high dimensionality, most of the volume of a sphere is concentrated in a *thin shell near the surface*.
- Distance functions losing their *usefulness* (for the nearest-neighbor criterion in feature-comparison algorithms, for example) in high dimensions.

Any Questions?





University of
Nottingham

UK | CHINA | MALAYSIA

QUIZ!

Question

In a medical test for a rare disease, out of 1000 people tested, 50 actually have the disease while 950 do not. The test correctly identifies 40 out of the 50 people with the disease as positive, but also wrongly identifies 30 of the healthy individuals as positive.

What is the precision of the test for this rare disease?

A) 96%

B) 92%

C) 80%

D) 57%

Answer

In a medical test for a rare disease, out of 1000 people tested, 50 actually have the disease while 950 do not. The test correctly identifies 40 out of the 50 people with the disease as positive, but also wrongly identifies 30 of the healthy individuals as positive.

What is the precision of the test for this rare disease?

A) 96%

B) 92%

C) 80%

D) 57%

Solution

In a medical test for a rare disease, out of 1000 people tested, 50 actually have the disease while 950 do not. The test correctly identifies 40 out of the 50 people with the disease as positive, but also wrongly identifies 30 of the healthy individuals as positive.

What is the precision of the test for this rare disease?

$$\text{Precision} = 40 / (40 + 30) = 57\%$$